Retrieve factors and order of operations

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ________________

1. Read the model: Find the HCF and LCM of 24 and 36. Copy or complete the first step.

2. Calculate 100 - 6 x (8 + 3).

3. Is 221 prime?

4. Miro says: Miro assumes brackets always make an expression larger. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the HCF and LCM of 24 and 36. Copy or complete the first step.
   Answer HCF 12; LCM 72.

2. Calculate 100 - 6 x (8 + 3).
   Answer 34.

3. Is 221 prime?
   Answer No, 221 = 13 x 17.

4. Miro says: Miro assumes brackets always make an expression larger. Is this correct? Explain.
   Answer Brackets change the order. The result may increase, decrease or stay the same.

Retrieve factors and order of operations

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ______________

1. Find the HCF and LCM of 24 and 36.

2. Calculate 100 - 6 x (8 + 3).

3. Is 221 prime?

4. Solve this independently and show every stage: Find the HCF and LCM of 24 and 36.

5. Use a second method or representation for: Calculate 100 - 6 x (8 + 3).

6. Check this result using an inverse, estimate or known fact: Is 221 prime?

7. Insert brackets and operations between 8, 6, 4 and 2 to make 40.

8. Identify and correct this misconception: Miro assumes brackets always make an expression larger.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Is 221 prime?

Teacher answers and checking prompts

1. Find the HCF and LCM of 24 and 36.
Answer HCF 12; LCM 72.

2. Calculate 100 - 6 x (8 + 3).
Answer 34.

3. Is 221 prime?
Answer No, 221 = 13 x 17.

4. Solve this independently and show every stage: Find the HCF and LCM of 24 and 36.
Answer HCF 12; LCM 72.

5. Use a second method or representation for: Calculate 100 - 6 x (8 + 3).
Answer Expected result: 34. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Is 221 prime?
Answer Expected result: No, 221 = 13 x 17. A valid check must be shown.

7. Insert brackets and operations between 8, 6, 4 and 2 to make 40.
Answer 8 x 6 - 4 x 2 = 40. Other correct expressions are acceptable.

8. Identify and correct this misconception: Miro assumes brackets always make an expression larger.
Answer Brackets change the order. The result may increase, decrease or stay the same.

9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Is 221 prime?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Retrieve factors and order of operations

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ______________

1. Insert brackets and operations between 8, 6, 4 and 2 to make 40.

2. Prove the answer to this question, rather than only calculating it: Find the HCF and LCM of 24 and 36.

3. Solve this in two different ways and compare them: Calculate 100 - 6 x (8 + 3).

4. Work backwards from the answer to reconstruct the question: No, 221 = 13 x 17.

5. Explain why this reasoning fails: Miro assumes brackets always make an expression larger.

6. Create a new example that tests the same idea as: Is 221 prime?

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Insert brackets and operations between 8, 6, 4 and 2 to make 40.
   Answer 8 x 6 - 4 x 2 = 40. Other correct expressions are acceptable.

2. Prove the answer to this question, rather than only calculating it: Find the HCF and LCM of 24 and 36.
   Answer Expected result: HCF 12; LCM 72. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Calculate 100 - 6 x (8 + 3).
   Answer Expected result: 34. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: No, 221 = 13 x 17.
   Answer One valid reconstruction is: Is 221 prime? Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro assumes brackets always make an expression larger.
   Answer Brackets change the order. The result may increase, decrease or stay the same.

6. Create a new example that tests the same idea as: Is 221 prime?
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.